Raha Kariminia

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Education

BSc in Computer Science, **Amirkabir University of Technology (Tehran Polytechnic)**, Tehran, Iran — Sep 2022 – Sep 2026

- **Cumulative GPA: 19.52/20**, on the U.S. Scale is **3.97/4**.
- **Major GPA: 20/20**, on the U.S. Scale is **4/4**.
- **Ranked 4th** in the 2022 Computer Science bachelor program.

Relevant Courses: AI and Workshop, Introduction to Neural Networks and Deep Learning, Graph Theory (all **20/20**), Data Mining (**18.34/20**).

Research Interests

- **Federated Learning:** personalized and heterogeneous federated learning
- **Graph ML:** GNNs, graph representation learning, node generation
- **Generative AI:** GANs, VAEs, diffusion transformers
- **Optimization:** MILP, combinatorial optimization, learning-augmented optimization
- **Other ML interests:** reinforcement learning, game theory, behavioral pattern mining

Awards & Certifications

Awards

- **Silver Medal**, **39th National Mathematical Olympiad**, Iran — Academic year 2020 – 2021

Certifications

- **Research Internship**, PURE Internship (Program for Undergraduate Research), Sabancı University — Jul 2026 – Aug 2026
  Conducted a research project titled "Learning-Augmented Optimization for Time-Dependent Soft-Clustered Vehicle Routing".
- **Algorithm Arena Course**, Amirkabir University of Technology — Sep 2023
  Earned skills in competitive algorithms through a skill-based course, covering graph algorithms, dynamic programming, greedy methods, and advanced data structures.
- **Scrum/Agile Workshop**, Ultima Training Tech Co. — Dec 2024
  Gained knowledge of Agile working methods and the Scrum framework and their differences through a workshop organized by this company.

Experience

Research collaboration with NORC lab, Amirkabir University of Technology — Apr 2026 – Sep 2026
Supervisor: Dr. Mehdi Ghatee

- Identified a key limitation in **federated graph learning** arising from **heterogeneity in feature domains** and **missing cross-subgraph neighbors** for each client. These factors result in **local bias** that cannot be handled by standard federated learning.
- Built **FedHyPr-MNG**, a hybrid framework that improves the **personalization–generalization trade-off** in federated graph learning by combining **global node generation, cross-client prototype sharing, and a dual-encoder design**. On our benchmarks, it outperformed baseline combinations of these components.
- This project started as my **bachelor's thesis**, which was completed in Sep 2026, and more advanced versions are still under development with my supervisor.

Summer Internship in PURE Program, Sabanci University — Jul 2026 – Aug 2026
Supervisor: Dr. Erhun Kundakcioglu

- Worked on a **learning-augmented optimization** project for a **time-dependent vehicle routing problem** in last-mile delivery, where customers are grouped into **flexible clusters** and served within **time windows**.
- Formulated a **MILP model** with **cluster-level** and **multi-commodity** constraints, solved it with **Gurobi**, and generated synthetic urban delivery datasets.
- Compared the MILP against a **DFS-Greedy** baseline that assigns clusters to vehicles via depth-first search and then builds routes with greedy nearest-neighbor and **2-opt local search**, cutting costs by **7.6–15.5%**. The internship ended in Aug 2026, and the work continues with my supervisor, with more advanced versions under development and learning-based extensions planned.

Undergraduate Teaching Assistant, Amirkabir University of Technology

- Advanced Artificial Intelligence (**Undergraduate TA** for **M.Sc. Level Course**) — Spring 2026
- Data Mining — Spring 2026
- Artificial Intelligence & Workshop — Fall 2024

- Foundation of Combinatorics — Fall 2023 & Spring 2023
- Numerical Linear Algebra — Fall 2025 & Spring 2026
- Basic Programming — Spring 2025
- Foundation of Numerical Analysis — Spring 2026

Projects

- **Autodiff Engine** — December 2025
  Python
  Built a reverse-mode autodiff engine from scratch for computing gradients in neural network training.
- **ADHD Mental Health Analysis** — June 2024
  Python, Scikit-learn, pandas
  Preprocessed mental health survey data and looked for patterns related to ADHD and other conditions using PCA with KMeans and classifier benchmarking.
- **Graph Learning on Zachary's Karate Club** — September 2024
  Python, NetworkX, Node2Vec, Scikit-learn
  Edge prediction with Adamic-Adar on clustered Node2Vec embeddings with KMeans at 0.97 purity.
- **Handwritten Letter Classification via QR Decomposition** — April 2025
  Python, NumPy, SciPy, EMNIST dataset
  this project implements Householder and Givens QR from scratch and an incremental Givens update for classification of letters.
- **PageRank: Three Implementations** — March 2025
  Python, NumPy, SciPy, NetworkX
  Compared eigendecomposition, power iteration, and NetworkX PageRank on web-BerkStan and found power iteration more accurate and also faster.

Skills

- **Languages:** Python, C/C++, LaTeX
- **ML & Data:** PyTorch, TensorFlow, Scikit-learn, NumPy, pandas, SciPy, Matplotlib
- **Graphs & Optimization:** NetworkX, Node2Vec, Gurobi, Pyomo, MILP
- **Backend:** FastAPI, PostgreSQL, SQLAlchemy, REST APIs, Docker, Git
- **Web:** React, TypeScript, JavaScript, HTML/CSS, Tailwind
- **Tools:** Poetry, Postman, VS Code, PyCharm, Visual Studio